

Supplemental Materials for *Environmental Belief System Networks and Pro-Environmental Behavior*

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S1. Dimensionality of the Study 1 NEP and CNS scales

I use exploratory graph analysis (EGA) and unique variable analysis (UVA) to check that the New Ecological Paradigm (NEP) and Connectedness to Nature Scale (CNS) items each form a single dimension before the scales enter the belief network. EGA estimates a regularized partial-correlation network over the items and detects communities with the Walktrap algorithm. A bootstrap with 500 resamples assesses the stability of that structure. UVA flags locally redundant item pairs with weighted topological overlap (wTO), using a redundancy threshold of 0.25.

For both scales the EGA solution splits into two communities that separate the forward-worded from the reverse-worded items rather than distinct content facets (Figure S1; Table S1, Table S2). This is a wording artifact rather than substantive multidimensionality, so I treat each scale as one dimension. UVA flags no item pairs above the 0.25 threshold (Table S3), so I do not drop or combine any items. A single-dimension EGA network score and the unit-weighted item mean correlate above .97 for each scale and relate to the behavior measures almost identically (Table S4), so the analysis in the main text uses the item means.

Table S1: EGA dimensionality and bootstrap (500 resamples) stability for the NEP and CNS item sets. Structural consistency is reported for each detected community.

Scale	Items	al-pha	EGA dimensions	bootEGA median dim	bootEGA modal dim	Structural consistency	Mean item stability
NEP	15	0.82	2	2	2 (85% of resamples)	0.75 / 0.78	0.96
CNS	14	0.81	2	2	2 (68% of resamples)	0.59 / 0.93	0.88

Figure S1: Exploratory graph analysis of the NEP and CNS items, separately and jointly. Nodes are items, edges are regularized partial correlations, and node color is the detected Walktrap community. Communities separate forward-worded from reverse-worded items.

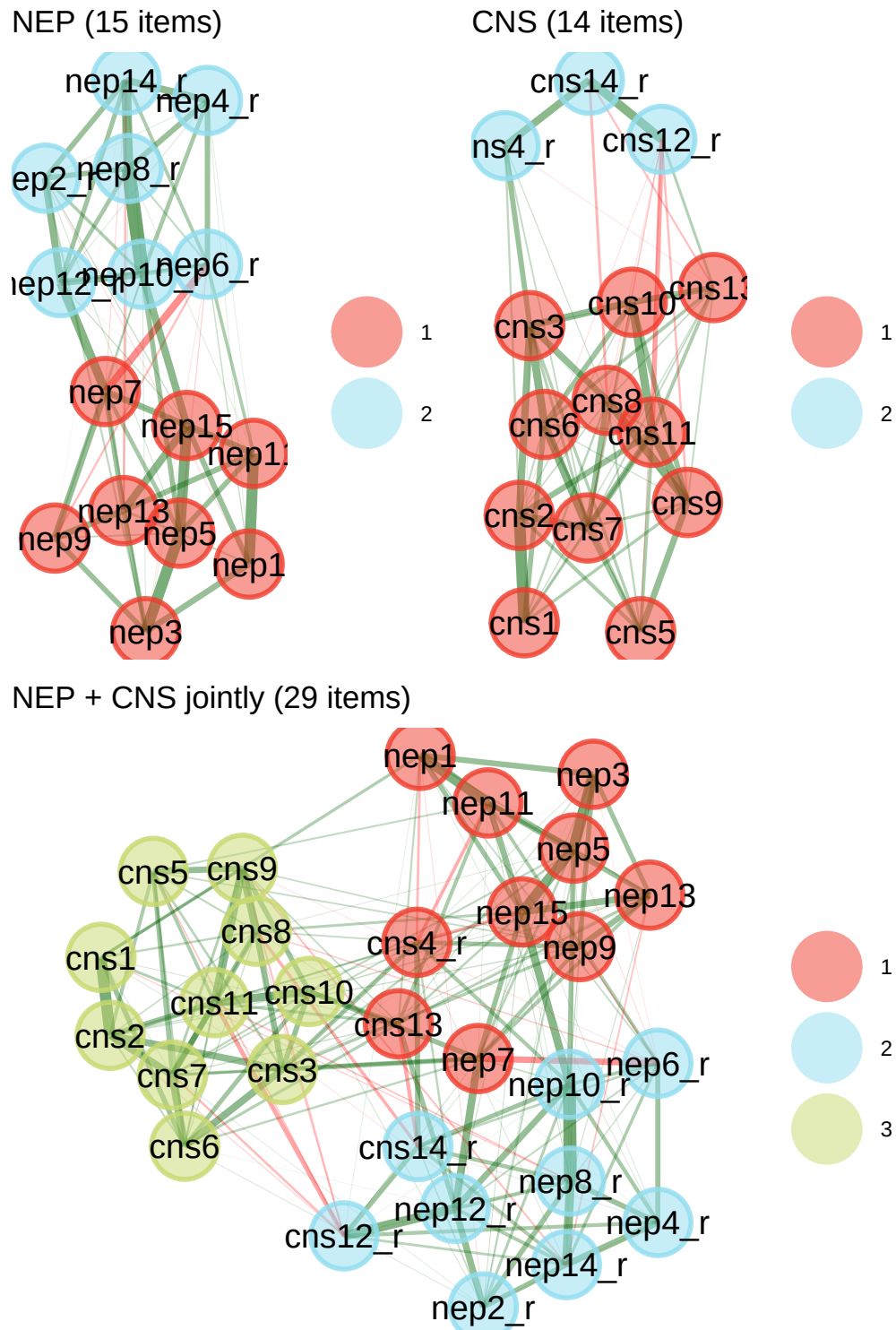


Table S2: Item-to-community assignment for the per-scale and joint NEP and CNS EGA solutions, with bootstrap item stability, by scale and item keying.

Item	Scale	Keying	Community (scale EGA)	Community (joint EGA)	Item stability
nep1	NEP	forward	1	1	0.994
nep2_r	NEP	reverse	2	2	1.000
nep3	NEP	forward	1	1	1.000
nep4_r	NEP	reverse	2	2	1.000
nep5	NEP	forward	1	1	1.000
nep6_r	NEP	reverse	2	2	0.778
nep7	NEP	forward	1	1	0.756
nep8_r	NEP	reverse	2	2	1.000
nep9	NEP	forward	1	1	0.940
nep10_r	NEP	reverse	2	2	1.000
nep11	NEP	forward	1	1	0.994
nep12_r	NEP	reverse	2	2	1.000
nep13	NEP	forward	1	1	0.990
nep14_r	NEP	reverse	2	2	1.000
nep15	NEP	forward	1	1	1.000
cns1	CNS	forward	1	3	0.866
cns2	CNS	forward	1	3	0.862
cns3	CNS	forward	1	3	0.902
cns4_r	CNS	reverse	2	1	0.876
cns5	CNS	forward	1	3	0.928
cns6	CNS	forward	1	3	0.900
cns7	CNS	forward	1	3	0.940
cns8	CNS	forward	1	3	0.820
cns9	CNS	forward	1	3	0.884
cns10	CNS	forward	1	3	0.930
cns11	CNS	forward	1	3	0.894
cns12_r	CNS	reverse	2	2	0.906
cns13	CNS	forward	1	1	0.706
cns14_r	CNS	reverse	2	2	0.954

Table S3: The five largest weighted topological overlap (wTO) values per scale from unique variable analysis. No pair reaches the wTO = 0.25 redundancy threshold.

Scale	Item i	Item j	wTO
NEP	nep8_r	nep10_r	0.230
NEP	nep1	nep11	0.214
NEP	nep3	nep5	0.195

Table S3: The five largest weighted topological overlap (wTO) values per scale from unique variable analysis. No pair reaches the wTO = 0.25 redundancy threshold.

Scale	Item i	Item j	wTO
NEP	nep8_r	nep14_r	0.185
NEP	nep5	nep15	0.185
CNS	cns1	cns2	0.217
CNS	cns12_r	cns14_r	0.165
CNS	cns9	cns11	0.163
CNS	cns10	cns11	0.156
CNS	cns6	cns7	0.145

Table S4: Correlations among the item mean, the single-dimension EGA network score, the UVA-reduced mean for the NEP, and the public and private behavior measures. Scores are oriented so that higher values indicate more pro-environmental positions.

	NEP_raw	NEP_EGA	NEP_UVA	CNS_raw	CNS_EGA	public	private
NEP_raw	1.000	0.992	0.987	0.447	0.372	0.170	0.304
NEP_EGA	0.992	1.000	0.980	0.465	0.397	0.190	0.320
NEP_UVA	0.987	0.980	1.000	0.469	0.398	0.178	0.310
CNS_raw	0.447	0.465	0.469	1.000	0.973	0.356	0.428
CNS_EGA	0.372	0.397	0.398	0.973	1.000	0.381	0.422
public	0.170	0.190	0.178	0.356	0.381	1.000	0.333
private	0.304	0.320	0.310	0.428	0.422	0.333	1.000

S2. Construction of the Study 2 belief scales

The ISSP environment module does not field the NEP or CNS scales, so I construct comparable belief measures from the module's attitude items. I start from 28 items covering concern, climate-change attribution, environment-economy tradeoffs, personal efficacy and commitment, the perceived danger of specific hazards, and enjoyment of nature. I orient every item so that higher values indicate more pro-environmental positions and then use exploratory graph analysis to identify the belief dimensions among them.

I compare candidate groupings with the Total Entropy Fit Index, where lower values indicate better fit (Table S5). The theoretical grouping by survey battery fits worst. A Louvain community solution over the full item pool fits best and recovers 5 communities, and dropping items does not improve fit. A bootstrap with 500 resamples recovers this solution in 95 percent of samples, and the structural consistency of each community is at least 0.95.

The survey battery grouping assigns each item to the part of the questionnaire in which it was asked. Concern is its own group, the climate change item joins the items about the environment and the economy, and the remaining groups are willingness to make sacrifices for the environment, personal efficacy and commitment, the danger of specific hazards, and enjoyment of nature. This six-group structure has a TEFI of -21.0, compared with -31.4 for the Louvain solution, so responses do not sort cleanly by where the items appear in the questionnaire. The hazard items and the two nature items hold together as their own communities. However, the items about willingness to pay higher prices, pay higher taxes, and cut one's standard of living join concern, doing what is right even when it costs more, and the view that environmental problems affect everyday life to form an environmental commitment community. The remaining sacrifice item, accepting a reduction in protected nature areas for economic development, joins most of the environment and economy items and most of the efficacy and commitment items in an ecological worldview community.

Table S6 lists the communities. Two survey batteries, the agree-disagree items about the environment and the economy and the items about personal efficacy and commitment, each split into a pro-environmental cluster and a skeptical cluster along the wording of the items rather than by content. This is the same wording artifact found for the NEP and CNS scales in Study 1. I therefore fold the small pro-environmental cluster into the larger ecological worldview community, which yields four scales. Table S7 gives the items and reliability of each scale and Table S8 the correlations among them and with a left-right ideology measure.

In the manuscript I divide the environmental commitment scale into three measures. The willingness items, paying much higher prices, paying much higher taxes, and accepting cuts in one's standard of living to protect the environment, along with doing what is right even when it costs more money and time, describe an intention to act rather than a belief about the environment. Willingness to sacrifice is the mean of the four willingness items ($\alpha = 0.78$). The remaining two items, concern about environmental issues and the view that environmental problems have a direct effect on everyday life, are only modestly related ($\alpha = .41$ as a pair), so each enters the network as a single-item node. The other three scales are unchanged.

For the behavior side, exploratory graph analysis of the six ISSP behavior items recovers 2 communities that match the public and private distinction used in Study 1. Recycling and product avoidance form one community, and group membership, petitions, donations, and protest form the other.

Table S5: Fit of candidate item groupings for the Study 2 belief items (pooled ISSP sample). TEFI is the Total Entropy Fit Index, where lower values indicate better fit.

Grouping	Dimensions	TEFI
a-priori batteries (crosswalk s7)	6	-21.0
EGA walktrap, full pool	3	-26.2
EGA louvain, full pool (chosen)	5	-31.4
EGA.fit optimized, full pool	5	-31.4
EGA louvain, drop v17	5	-30.4
EGA louvain, drop v17/v24/v35	4	-26.2
EGA louvain, drop nature (v46/v47)	4	-27.7
EGA louvain, US only	5	-21.2

Table S6: The five communities recovered by exploratory graph analysis of the Study 2 belief items.

Community	Label	Items
1	Environmental commitment	v15, v26, v27, v28, v31, v36
2	Ecological worldview (skeptical wording)	v17, v20, v21, v23, v24, v29, v30, v32, v33, v34, v35
3	Growth harms the environment	v22, v25
4	Perceived hazard danger	v37, v38, v39, v40, v41, v42, v43
5	Nature affinity	v46, v47

Table S7: The four Study 2 belief scales, their items, and reliability (pooled ISSP sample). The two-item growth-harms community is folded into the ecological worldview scale.

Scale	Name	Items	Alpha
Environmental commitment	envcom	v15, v26, v27, v28, v31, v36	0.75
Ecological worldview	world-view	v17, v20, v21, v23, v24, v29, v30, v32, v33, v34, v35, v22, v25	0.76
Perceived threat	threat	v37, v38, v39, v40, v41, v42, v43	0.80
Nature affinity	nature	v46, v47	0.61

Table S8: Correlations among the Study 2 belief scales, a left-right ideology measure, and the six behavior items (pooled ISSP sample). Higher values on every measure indicate more pro-environmental positions or more frequent behavior.

	en- vcom	world- view	threat	na- ture	PARTY_LR	recy- cle	avoid_buy	group_mem- ber	peti- tion	do- nate	protest
envcom	1.00	0.34	0.28	0.15	-0.22	0.20	0.36	0.18	0.26	0.25	0.17
world- view	0.34	1.00	0.30	0.34	-0.23	0.33	0.27	0.10	0.34	0.25	0.14
threat	0.28	0.30	1.00	0.07	-0.22	0.05	0.18	0.04	0.14	0.06	0.07
nature	0.15	0.34	0.07	1.00	-0.03	0.34	0.22	0.09	0.23	0.19	0.10
PARTY_LR	-0.22	-0.23	-0.22	-0.03	1.00	-0.03	-0.09	-0.10	-0.20	-0.13	-0.14
recycle	0.20	0.33	0.05	0.34	-0.03	1.00	0.37	0.05	0.20	0.16	0.05
avoid_buy	0.36	0.27	0.18	0.22	-0.09	0.37	1.00	0.14	0.23	0.21	0.13
group_mem- ber	-0.18	0.10	0.04	0.09	-0.10	0.05	0.14	1.00	0.27	0.34	0.23
petition	0.26	0.34	0.14	0.23	-0.20	0.20	0.23	0.27	1.00	0.40	0.32
donate	0.25	0.25	0.06	0.19	-0.13	0.16	0.21	0.34	0.40	1.00	0.24
protest	0.17	0.14	0.07	0.10	-0.14	0.05	0.13	0.23	0.32	0.24	1.00

S3. Belief-to-behavior distances by country

Table S9 reports, for each country, the belief with the shortest mean path to the behavior set and its 95 percent bootstrap interval, along with the proportion of 1,000 resamples in which that belief is the closest. A support value near 1 means the ordering is stable, and a value near one ninth, with nine beliefs, means it is close to a coin flip. Where two beliefs tie for closest, the table shows the one with more bootstrap support.

Table S9: For each ISSP country, the belief with the shortest mean path to the behavior set, its point estimate with a 95 percent bootstrap interval, and the proportion of 1,000 resamples supporting it as the closest belief.

Country	Belief	Distance	Support
AT	Willingness to sacrifice	5.8 [4.6, 7.7]	0.77
AU	Environmental concern	7.2 [5.0, 9.6]	0.55
CH	Willingness to sacrifice	6.7 [5.7, 8.2]	0.97
CN	Willingness to sacrifice	6.8 [5.7, 8.6]	0.91
DE	Willingness to sacrifice	7.6 [6.0, 10.2]	0.74
DK	Willingness to sacrifice	6.9 [5.0, 7.7]	0.64
ES	Environmental concern	6.7 [5.6, 8.4]	0.95
FI	Willingness to sacrifice	6.4 [5.7, 8.1]	0.55
FR	Environmental concern	6.2 [5.0, 8.4]	0.76
HR	Environmental concern	9.6 [6.8, 12.7]	0.25
HU	Willingness to sacrifice	4.0 [3.4, 5.7]	1.00
IN	Nature affinity	8.1 [5.7, 10.5]	0.96
IS	Nature affinity	6.9 [5.5, 9.5]	0.66
IT	Willingness to sacrifice	4.8 [3.7, 6.9]	0.98
JP	Willingness to sacrifice	6.0 [4.5, 9.4]	0.92
KR	Willingness to sacrifice	9.3 [6.0, 13.4]	0.82
LT	Willingness to sacrifice	5.6 [4.5, 7.8]	0.89
NO	Willingness to sacrifice	8.6 [6.0, 10.2]	0.55
NZ	Environmental concern	6.5 [4.9, 9.9]	0.64
PH	Nature affinity	15.5 [8.2, 27.4]	0.64
RU	Willingness to sacrifice	6.6 [5.3, 8.5]	0.95
SE	Willingness to sacrifice	6.9 [5.7, 9.2]	0.59
SI	Environmental concern	7.1 [5.1, 10.7]	0.75
SK	Willingness to sacrifice	6.8 [5.1, 10.6]	0.47
TH	Environmental concern	6.2 [4.6, 8.5]	0.84
TW	Ecological worldview	8.7 [6.6, 11.8]	0.33
US	Nature affinity	6.4 [4.7, 7.3]	0.83
ZA	Willingness to sacrifice	5.1 [4.4, 5.9]	0.98

References